MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of Faculty: Bhanwar Veer Singh
Subject & Subject Code: Engineering Chemistry (1FY2-03)
Semester: I **Session: 2022-23 (Odd Sem.)**
Branch: All **Batch: -C and D**
Unit No.: 1 **Date of Submission: 27.9.22**

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References:

1. Engineering Chemistry,by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

Faculty **HOD** **Dean Academics**

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Lecture No.	Topics Covered	Plagiarism(%)	Unique (%)
1	Common impurities, hardness, Degree of hardness, Units of hardness	9	91
2	Determination of hardness by complexometric (EDTA method),	7	93
3	Municipal water supply: Requisite of drinking water, Purification of water; sedimentation	12	88
4	Filtration, disinfection, breakpoint Chlorination	4	96
5	Boiler toubles: Scale & Sludge formation	17	83
6	Priming and Foaming,Boiler corrosion and Caustic embrittlement	25	75
7	Water Softening : Lime- Soda process	31	69
8	Zeolite (Permutit) process	12	88
9	Demineralization process	30	70
10	Numerical problems based on Hardness, EDTA, Lime-Soda and Zeolite Process	56	44

References:

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UNIT-1 WATER

Introduction:

The Planet earth is abundantly rich in the water, which constitute 75% of its surface. Besides being as essential ingredient of animal and plant life, water forms the basis of human life. It constitutes about 65% of human body. Water is essential not only for survival of human beings, animals, plants and all other living beings but also utilized in a large amount in industries. Out of the total water consumed by human beings, more than 50% of its consumed for industrial activity and only a small proportion is used for drinking water. The various sources of water found in planet earth are Surface water, rain water and underground water etc.

Common Impurities in Water, Hardness, Degree of Hardness, Unit of Hardness

Types of Impurities in Water

Dissolved Impurities: It may be due to presence of dissolved salts like carbonates, bicarbonates, sulphates and chloride of calcium, magnesium, iron, sodium and potassium.

Gases: CO_2, O_2, N_2, H_2S, NH_3 etc.

Suspended Impurities: These impurities impart turbidity, colour and odour to water example- Inorganic Sand Clay.

Colloidal Impurities: Vegetable and animal matter, silica and clay, organic waste products, coloring matter etc.

Micro Organism: Bacteria, Algae, Fungi, Virus etc.

Hardness of Water: "**The characteristic properties of water which prevent the lathering of soap,** due to presence of certain salts of calcium, magnesium and other heavy metals (like Al^{+3}, $Fe^{+3}$and Mn^{+2}) dissolved in water".

It was originally define as ***"the soap consuming capacity of a sample of water".***

Water can be classified in to two categories:

1. **Hard Water:** Water which does not produce lather with soap solution readily is called hard water. e.g. sea water, river water.
2. **Soft Water:** Water which produce lather with soap solution readily is called soft water. e.g. rain

water.

Cause of Hardness: Hardness of water is due to the presence of calcium and magnesium salts generally, soap consist of sodium salt of higher fatty acids such as stearic, oleic and palmitic, which are highly soluble in water and thus exert their cleaning action. If calcium and magnesium salts are present in water, these react soluble sodium soap to form insoluble salt of calcium and magnesium.

$$2\,C_{17}H_{35}COONa + CaSO_4 \longrightarrow (C_{17}H_{35}COO)_2Ca + Na_2SO_4$$

(Sodium Stearate) **(Calcium Stearate)**

(Soluble salt) **(Insoluble salt) ppt**

$$2C_{17}H_{35}COONa + MgCl_2 \longrightarrow (C_{17}H_{35}COO)_2\,Mg + NaCl$$

(Sodium Stearate) **(Magnesium Stearate)**

(Insoluble salt) ppt

The white scum or precipitate of insoluble soap of calcium and magnesium is formed. Lather is not produced until the cations, Ca^{+2} and Mg^{+2} are completely precipitate out in the form of insoluble salts**.**

Type of Hardness: Hardness of water are classified into two type.

1. **Temporary or Carbonate hardness:** This type of hardness caused due to the presence of bicarbonates and carbonates of calcium and magnesium in water and can be removed simply by boiling.

$$Ca(HCO_3)_2 \xrightarrow{\text{boil}} CaCO_3 + H_2O + CO_2$$

$$Mg(HCO_3)_2 \xrightarrow{\text{boil}} MgCO_3 + H_2O + CO_2$$

2. **Permanent or Non-Carbonate hardness:** It is due to the presence of sulphates and chlorides of Ca, Mg, and other heavy metals and cannot be removed by boiling. It is removed by the use of chemical agents.

Degree of Hardness: The degree of hardness of water is expressed in term of equivalent amount of $CaCO_3$. Although hardness of water is never present in the form of calcium carbonate because it is insoluble in water and hardness is actually caused by bicarbonates, chlorides and sulphates of calcium and magnesium.

$CaCO_3$ as the standard for reporting hardness of water is due to its molecular weight which is exactly 100 and equivalent weight is 50. The equivalent of $CaCO_3$ for hardness producing substance can be calculated

Equivalent of $CaCO_3$ = Weight of h.p.s × equivalent weight of $CaCO_3$/ Equivalent weight of h.p.s

Or

Equivalent of $CaCO_3$ = M.F × Weight of h.p.s

Where : h.p.s = hardness producing substance
M.F = Multiplication Factor

Multiplication factor for various dissolved salt:-

Dissolved salt ion	Molar mass	chemical equivalent	Multiplication factor
$Ca(HCO3)_2$	162	81	100/162
$Mg(HCO3)_2$	146	73	100146
CaCO3	100	50	100/100
MgCO3	84	42	100/84
CaSO4	136	68	100/136

Unit of Hardness: Unit of hardness of water is expressed in following concentration terms.

1. **Parts Per million (ppm):** It is a part of $CaCO_3$ equivalent hardness per 10^6 parts of water.
2. **Milligram per liter (mg/liter):** It is defined as the number of milligram of $CaCO_3$ equivalent hardness present in one liter of water that is :

   1 mg/lit = 1 mg of CaCO3 equivalent hardness in one liter of water

   1 mg/lit = 1 ppm
3. **Degree Clarke (ºCl):** It is define as number of grains (64.8mg) of $CaCO_3$ equivalent hardness per gallon (4.55L) of water. i.e. a parts of $CaCO_3$ equivalent hardness per 70000 part of water.

   1 ºCl = one grains of $CaCO_3$ per 70000 part of water
4. **Degree French (ºFr) :** It is a part of $CaCO_3$ equivalent hardness per lac (10^5) part of water that is :

   1 ºFr = 1 part of $CaCO_3$ equivalent hardness per 10^5 parts of water

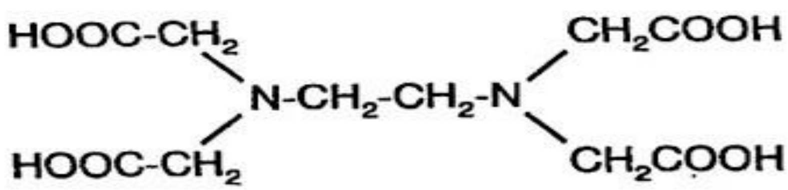
References:

1. Engineering Chemistry,by S. Rattan
2. Engineering Chemistry, Wiley India

Determination of hardness by complexometric method (EDTA method) :

Determination of hardness by complexometric method (EDTA method) :

This method is used for the determination of hardness in sample water. Ethylene diamine tetra acetic acid (EDTA) is a strong complexing agent. The structure of EDTA is:



EDTA acts as a tetra dentate and hexadentate ligands and binds with calcium and magnesium ions. It is insoluble in water and used as a disodium salt which is quite soluble in water. It forms a very stable, colorless, 1:1 complex with calcium and magnesium ions.

Complexing property of EDTA is used to estimate the amount of hardness causing ions in a water, for this purpose the water sample buffered at pH= 9-10 is titrated against EDTA using **Erichrome black-T (EBT)** indicator. Erichrome black-T indicator is a blue colored azo dye which is capable of forming soluble wine red coloured complex with Ca^{+2}, Mg^{+2}.

Procedure:

Preparation of EDTA Solution- Dissolve 3.72 gm disodium salt of EDTA in distilled water make up to 1 liter.

Preparation of indicator (EBT) - 0.5 gm of Eriochrome black-T is dissolve in 100 ml ethanol. EBT is blue dye.

Preparation of Buffer solution- 70 gm of NH_4Cl is dissolve in 570 ml of NH_4OH and the solution is diluted with distilled water to make 1 liter.

Preparation of Standard hard water- Take 1 gm of dry $CaCO_3$ then dissolve in minimum amount of dilute HCl and then evaporated to dryness on water bath then residue is dissolve in distill water and solution is made 1L by adding distill water. This standard hard water contains 1mg of $CaCO_3$ equivalents in 1 ml of this solution.

Titration-

Standardization of EDTA Solution- Fill up the burette with EDTA solution and pipette out 50 ml of standard hard water in to conical flask. Add 10 ml of buffer solution and 2-3 drops of EBT in conical flask. Titrate this solution against EDTA until the wine red color, change to blue (due to EBT).

Determination of total hardness of water: As per the same procedure disused above, titrate 50ml of unknown water sample against EDTA

Determination of Permanent hardness of water: Take 250ml of water sample in a 500ml beaker and boil it till the volume is reduced to about 50ml. this step causes all bi carbonates decomposed in to insoluble $CaCO_3$ and $Mg(OH)_2$ **.** Filter and wash the precipitate with distilled water and quantitatively collect the filtrate and wash in a 250ml conical flask and makeup the volume to 250ml with distilled water. Titrate 50ml of this water sample against EDTA.

Calculation-

Strength of EDTA Solution

1 mL of standard hard water contains = 1mg of $CaCO_3$

50 mL of standard hard water = 50mg of $CaCO_3$

50 mL of standard hard water = V_1 mL of EDTA

50 mg of $CaCO_3$ eq. consumed by = V_1 mL of EDTA

V_1 mL of EDTA consumed by = 50 mg of $CaCO_3$ eq.

1 mL of EDTA consumed by = $50/V_1$ mg of $CaCO_3$ equivalent

Total Hardness:

50mL of unknown sample (before boiling) of hard water consumed = V_2 mL of EDTA solution

1 ml of unknown sample of hard water consumed = $V_2/50$ mg of $CaCO_3$ equivalent

$50/V_1$ mg of $CaCO_3$ equivalent consumed = $50/V_1 \times V_2/50$ mg of $CaCO_3$ equivalent

1000 ml of sample of hard water consumed = $50/V_1 \times V_2 \times 1000/50$ mg of $CaCO_3$ equivalent

= $V_2/V_1 \times 1000$ mg of $CaCO_3$ equivalent

Total hardness = $V_2/V_1 \times 1000$ mg/Liter

Thus total hardness = $V_2/V_1 \times 1000$ppm

Permanent Hardness:

50mL of sample of hard water (after boiling) consumed = V_3 mL of EDTA solution

1ml of sample of hard water (after boiling) consumed = V3/50 mL of EDTA solution

$50/V_1$ mg of $CaCO_3$ equivalent consumed = $50/V_1 \times V3/50$ mg of $CaCO_3$ equivalent

1000 ml of sample of hard water (after boiling) consumed = $50/V_1 \times V3 \times 1000/50$ mg of $CaCO_3$

equivalent

$= V_3/V_1 \times 1000$ mg/Litre

Permanent hardness $\boldsymbol{= V_3/V_1 \times 1000}$**ppm**

Temporary hardness:

Temporary hardness = (Total hardness) - (Permanent hardness)

$= [(V_2/V_1 \times 1000) - (V_3/V_1 \times 1000)]$ mg of $CaCO_3$ equivalent

$=1000/V_1\ (V_2-V_3)$ mg/L

Thus temporary hardness = $\boldsymbol{1000((V_2-V_3)\ /V_1}$ **ppm**

Advantages of EDTA Method:

- Greater accuracy
- More rapid procedure
- Convenience

Disadvantage of hard water

- Hard water is not useful for various domestics' purpose, viz washing bathing drinking etc.
- Hard water is harmful for many industries such as textile, sugar, paper etc.
- Hard water is not suitable for steam raising in boilers, since the caustic embrittlement and the other problems caused by hard water.

References:

1. Engineering Chemistry,by S. Rattan

2. Engineering Chemistry, Wiley India

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L-3, Municipal water supply, Purification of water, sedimentation

Municipal water:

Water is a vital body fluid and is essential for all living creatures. Human beings consume a considerable amount of water for drinking and other domestic purposes. The principal requirement for water to be supplied for drinking purpose i.e., municipal water satisfy certain quality criteria with respect to appearance portability and health.

Requisites of Drinking water:

The common specifications for drinking water are given below.

Physical: It should be clear and odour less.

- Should be pleasant to taste and perfectly cool.
- Its turbidity should not exceed 10ppm.

Chemical: Should be free from harm full gases like H_2S.

- Its P^H should be in the range of 7.0-8.5.
- Chloride and Sulphate should be less than 250ppm.
- Total hardness and Total dissolved solids (TDS) should be less than 500 ppm.
- Fluoride content should be less than 1.5.

Biochemical: It should be free from pathogens.

Purification of water: Rivers, lakes and well are the most common source of water used by municipalities. The treatment method depends directly on the impurities present. For removing various types of impurities the following treatment processes are employed.

Process used for removal	Impurity
Screening	Floating matter such as leaves wood pieces etc.
Sedimentation	Suspended impurities such as clay , sand etc.
Coagulation	Fine suspended matter (salt)
Filtration	Micro-Organism and colloidal matter
Disinfection	Pathogenic bacteria

Screening: In this process removing floating materials like wood pieces, leaves etc. from water.

Sedimentation: The process of removal of suspended impurities by allowing the water to stand undisturbed for 2-6 hours in big tank is known as *sedimentation*.

Most of the suspended particles settle down at the bottom of tank due to force of gravity and they are removed. 70-75% of the suspended matter is removed during this process. The tank where the flow of water is retarded is called the setting or sedimentation tank.

Sedimentation and Coagulation: In this process, fine suspended and colloidal particle are removed by addition of requisite amount of chemical (as coagulants) to water before sedimentation. Coagulants like alum or ferrous sulphate produce Al^{+3}, Fe^{+3} ions which neutralized the oppositely charged colloidal and clay particles amount of chemicals. Some common chemical used as coagulating are as follows:

$$\underset{\text{(Coagulant)}}{Al_2(SO_4)_3} + 3Ca(HCO_3)_2 \longrightarrow 2Al(OH)_3\downarrow + 3CaSO_4 + 6CO_2$$

$$\underset{\text{SODIUM ALUMINATE}}{NaAlO_2} + 2H_2O \longrightarrow 2Al(OH)_3\downarrow + NaOH$$

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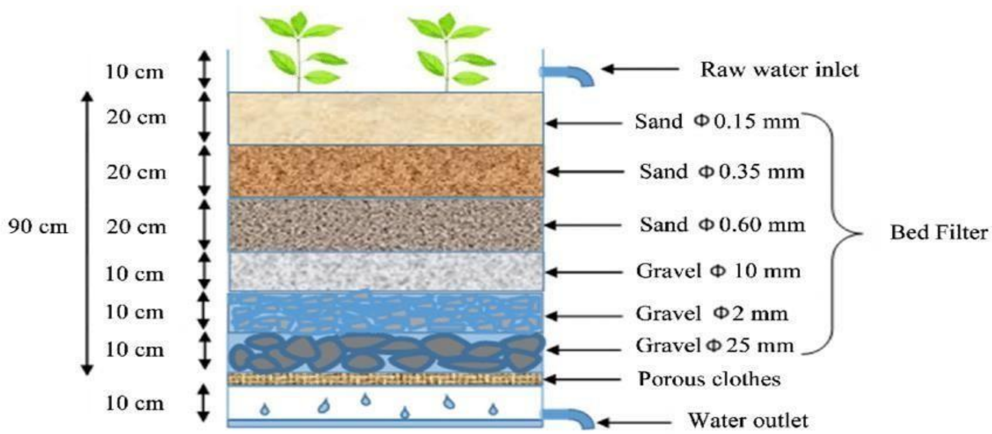
L-4, Filtration, Disinfection, Breakpoint Chlorination

Filtration: Filtration is the process of removing insoluble, colloidal and bacterial micro-organism etc. Filtration is carried out by sand filter.

A sand filter consists of tank with top layer of fine sand placed over coarse sand layer and gravels. During filtration, the sand pores get clogged due to retention of impurities in the process which can be removed by regeneration process.



SAND FILTER

Disinfection and Sterilization or Removal of micro organism: "The process of destroying killing the pathogenic micro-organism from water is known as disinfection." By this process removed the micro organism and make safe water for drinking.

Disinfectants generally used for water are:

a) Bleaching powder ($CaOCl_2$)

b) Chlorine

c) Chloramine ($ClNH_2$)

d) Ultraviolet rays

e) Ozone

f) $KMnO_4$

Disinfection by Chlorine or Bleaching Powder: Chlorine is a most common disinfectant in water treatment. Chlorine may be added in the form of bleaching powder and added to water its formed HOCl (hypochlorous acid) which act as a powerful germicide.

$CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$

$Cl_2 + H_2O \longrightarrow HCl + HOCl$

$HOCl + \text{Bacteria} \longrightarrow$ Bacteria are killed

Disinfection by Chloramine: Chloramine has better bactericidal action and more stable than chlorine.

$ClNH_2 + H_2O \longrightarrow HOCl + NH_3$

$Cl_2 + NH_3 \longrightarrow ClNH_2 + HCl$

(Chloramine)

Disinfection by UV Rays: UV radiation emitted from electric mercury vapour lamp is capable of disinfection of water. This process is particular useful for disinfection of swimming pool water. This is highly expensive.

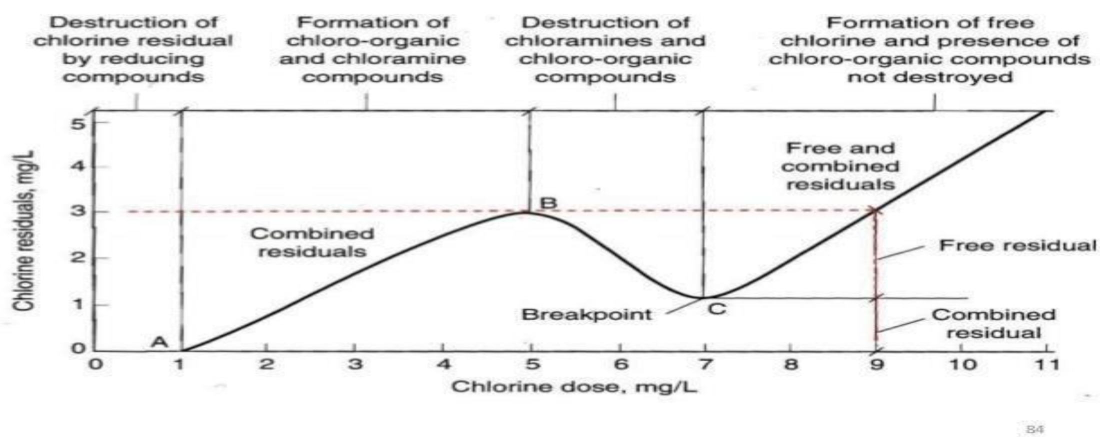
Disinfection by Ozone: Ozone is a powerful disinfectant and is absorbed by water. Ozone is highly unstable and decomposes to give nascent oxygen which is capable of destroying bacteria.

$O_3 \longrightarrow O_2 + [O]$ **(Nascent oxygen)**

By Potassium permanganate: $KMNO_4$ used for sterilization of water.

Breakpoint Chlorination:

Break point chlorination is the chlorination of water to such an extent that not only living organisms but also organic impurities present in water are completely destroyed.



***"Break point is the point at which free residual chlorine begins to appear"*. When chlorine is added to water, it is used for oxidation of reducing compound present, chlorination of organic impurities, free ammonia and disinfection of bacteria.**

Break Point Chlorination takes place in four stages;-

Stage I- In the first stage, chlorine used in very low dose and no free residual chlorine since all the added chlorine gets consumed for complete oxidation of reducing substance present in water.

Stage II- As the amount of chlorine increases there is steady increase in the amount of residual chlorine also. This stage corresponds to the formation of chloro-organic compound & chloramines.

Stage III- At still higher doses of applied chlorine, oxidation and destruction of organic compounds and micro organism sets in, consequently, the amount of free residual chlorine also decreases. When the oxidative destruction is complete, it reached a minima.

***Stage IV* -**After minima, the added chlorine is not used in any reaction. The residual chlorine keep increasing in direct proportion to added chlorine.

Hence, for effectively killing the micro-organisms, sufficient chlorine has to be added. Addition of chlorine in such dosage is known as ***Break Point Chlorination.***

References:

1. Engineering Chemistry,by S. Rattan

2. Engineering Chemistry, Wiley India

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L-5, Boiler troubles Scale and Sludge formation

Boiler troubles: Boiler troubles are the defects in boiler which are caused by imperfect water. Untreated water, containing impurities may lead to the following problems in boilers.

1. Scale and sludge formation
2. Boiler corrosion
3. Caustic Embrittlement
4. Priming and foaming

Scale and sludge formation: Boilers are used for steam generation. When hard water is evaporated, progressively the concentration of dissolved salt is increased.

When their saturation points are reached, the dissolved salts of calcium and magnesium along with other soluble impurities are precipitated on the inner walls of boilers and in due course of time adhere to the metal surface in the form of scales and sludges.

Sludge- When precipitation takes place in the form of loose, non adherent and slimy precipitate, it is called sludge.

Causes of sludge Formation:

Sludge is a soft, loose and slimy precipitate and formed by substances which have greater solubility in hot water than in cold water, e.g., $MgCO_3$, $MgCl_2$, $CaCl_2$, $MgSO_4$, etc.

Scale-If the precipitated matter forms a hard, adhering coating on the inner walls of the boiler, it is called scale.

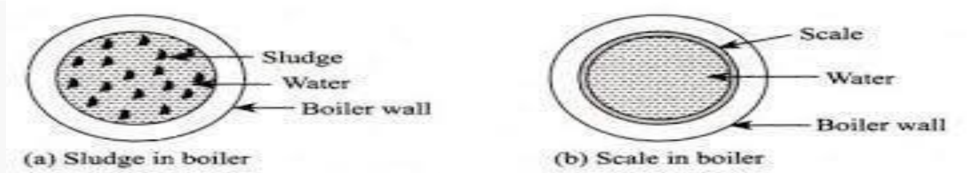
Causes of scale Formation:

1) *Decomposition of calcium bicarbonate*

$$\mathbf{Ca(HCO_3)_2 \rightarrow CaCO_3 \downarrow + H_2O + CO_2 \uparrow}$$

2) Deposition of calcium sulphate**:** calcium sulphate is soluble in cold water but solubility decreases with the raise of temperature. Consequently $CaSO_4$ gets precipitate as hard scale on the heated portions of the boiler.
3) ***Presence of silica:*** Even small quantity of silica result in the formation calcium silicates and magnesium silicates which are sparingly soluble in cold water but are completely insoluble in hot water.
4) ***Hydrolysis of magnesium salts:*** Dissolved magnesium salts undergo hydrolysis forming precipitate magnesium hydroxide which forms a soft type of scale.

$$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 \downarrow + 2HCl \uparrow$$



Sludges and scales in boiler

Disadvantages of sludge &scale formation:

- ✓ Sludge's are poor conductor of heat, so they tend to waste a portion of heat generated and thus decreases the efficiency of boiler.
- ✓ Excessive sludge formation disturbs the working of the boiler, causing even choking of the pipes.
- ✓ ***Wastage of fuel**:* Scales have a low thermal conductivity which decreases the amount of heat transferred to boilers. It results in the wastage of fuels.
- ✓ ***Begging:*** The distortion of boiler materials is known s begging. Due to overheating, rapid reaction between water and iron occurs at high temperature, causing thinning of the boiler materials.

Prevention of sludge formation:

(1) By using well softened water

(2) Blow-down operation - Removal of concentrated water and replacing it by fresh water is called blow down operation.

Removal of Scales: The scale can be prevented by-

1) Internal treatment

2) External treatment

1) <u>Internal treatment:</u> In this method the boiler feed water refers to the treatment of water in the boiler itself which involved addition of certain chemical directly to the water to remove the scale forming salt.

Some method used for internal treatment.

- ➢ **Colloidal Conditioning*:*** Scale formation can be reduced by introducing colloidal substance like kerosene, tanning agar etc. to the boiler water.
- ➢ **Carbonate Conditioning*:*** The scale forming salt like calcium is converted in to calcium

carbonate which can be removed easily.

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$$

- **Phosphate Conditioning*:*** The phosphate reacts with calcium and magnesium salt in the boiler water foaming easily removal soft sludges of calcium and magnesium phosphates.

$$3CaSO_4 + 2Na_3PO_4 \rightarrow \underset{\text{(Sludges)}}{Ca_3(PO_4)_2} + 3Na_2SO_4$$

- **Calgon conditioning** :- Calgon [sodium hexametaphosphate $Na_2[(Na_4PO_3)_6]$ is added to boiler water which prevents the scale formation by forming the soluble complex with calcium sulphate.

$$2CaSO_4 \text{ (Boiler water)} + Na_2[Na_4P_6O_{18}] \rightarrow Na_4[Ca_2P_6O_{18}] + 2Na_2SO_4$$

Highly soluble complex

Differences between the external and internal treatment methods

Sr No.	Internal Treatment	External Treatment
1	It is carried out in the boiler itself.	It is carried outside the boiler, before water enters the boiler.
2	It is required in low-pressure boilers	It is required in high-pressure boilers
3	It include colloidal conditioning, carbonate conditioning, phosphate conditioning, etc.	It includes zeolites process, lime process, lime soda process, and ion exchange process.

<u>**External treatment:**</u> This process done by following methods.

1. Lime-soda process
2. Zeolite or Permutit process
3. Ion exchange method or demineralization

Boiler corrosion, Caustic Embrittlement, Priming and Foaming

Boiler corrosion:

Boiler corrosion is decay or disintegration of boiler body material either due to chemical or electro-chemical reaction with its environment.

Disadvantage of Corrosion:

1) Shortening of boiler life

2) Leakage from the joint and rivets.

3) Increased cost of repairs and maintenance

Causes of boiler corrosion are:

1) ***Dissolved oxygen:*** Water usually contains about 8 ppm of dissolved oxygen at room temperature. Dissolved oxygen in water, in presence of high temperature, attacks boiler material.

$$2Fe + 2H_2O + O_2 \rightarrow 2\,Fe(OH)_2$$

$$4Fe(OH)_2 + O_2 \rightarrow 2(Fe_2O_3.2H_2O)$$

Removal of dissolved oxygen

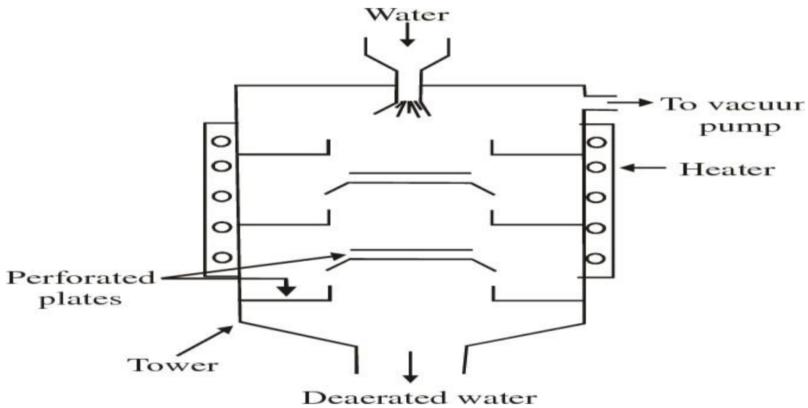
1) By adding the hydrazine or sodium sulphite or sodium sulphide Thus,

$$\underset{\text{(Hydrazine)}}{N_2H_4} + O_2 \rightarrow N_2 + 2H_2O$$

$$\underset{\text{(Sodium sulphite)}}{2Na_2SO_3} + O_2 \rightarrow 2\,Na_2SO_4$$

$$\underset{\text{(Sodium sulphide)}}{Na_2S} + 2O_2 \rightarrow Na_2SO_4$$

2) By mechanical de-aeration, in this process water spraying in a perforate plate- fitted tower. Removal of dissolved O_2 is ensured by applying high temperature and vacuum (according to Henry Law).



TOWER FOR MECHANICAL DEAERATION OF WATER

2) Dissolved carbon dioxide : There are two sources of CO_2 in boiler water viz, dissolved CO_2 in raw water and CO_2 formed by decomposition of bicarbonates in H_2O.

$$Mg(HCO_3)_2 \rightarrow MgCO_3 + H_2O + CO_2$$

Carbon dioxide in presence of water forms carbonic acid which has a corrosive effect on the boiler materials like any other acid.

$$H_2O + CO_2 \rightarrow H_2CO_3$$

Removal of CO_2:

(1) By mechanical deaeration along with O_2.

(2) By filtering water through limestone.

$$\mathbf{CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2}$$

(3) By addition of appropriate quantity of ammonium hydroxide

$$\mathbf{CO_2 + NH_4OH \rightarrow (NH_4)_2CO_3 + H_2O}$$

Caustic Embrittlement:

Caustic embrittlement is the phenomenon during which boiler materials becomes brittle due to accumulation of caustic substance. It takes place in high pressure boiler due to presence of highly alkaline water.

During softening process, calculated quantity of Na_2CO_3 is added which decomposed to give sodium hydroxide and CO_2 due to sodium hydroxide boiler water becomes "caustic".

$$\mathbf{Na_2CO_3 + H_2O \rightarrow 2NaOH + CO_2}$$

The caustic embrittlement can be explained by the formation of the conc. cell. The iron surrounded by concentrated NaOH becomes anodic side, which is corroded whereas the iron surrounded by dil. NaOH becomes cathodic side.

At high temp. in cracks and joints under stress, this concentrated NaOH dissolves iron and form Sodium ferrate.

$$\mathbf{Fe + 2NaOH \rightarrow 2\ Na_2FeO_2 + H_2O}$$

This sodium ferrate at high temp. decomposed into NaOH and Fe_3O_4 thus again NaOH regenerate which further dissolve iron.

$$\mathbf{2\ Na_2FeO_2 + 4H_2O \rightarrow 6NaOH + Fe_3O_4 + H_2O}$$

Prevention*:* Caustic embrittlement can be prevented by following-

- By using sodium phosphate as softening reagent instead ofsodium carbonates.
- By adding tannin or lignin to boiler water which blocks the cracks inthe boiler walls.
- By adding sodium sulphate to boiler water, Na_2SO_4 also blocks cracks thereby preventing deposition of caustic soda solution in these.

Priming and Foaming

Priming: When steam is produced rapidly in the boiler, some droplets of the liquid water are carried along with the steam. This process of wet steam formation is called "***Priming***".

Causes of Priming -

- The presence of considerable quantities of dissolved solids.
- Steam velocities high enough to carry droplets of water in to the steam pipe.
- Sudden boiling
- Faulty design of boiler.

Disadvantage of Priming and foaming

- Dissolved salts in boiler water are carried by the wet steam to super-heater and turbine blades, where they get deposited as water evaporates. This water reduces their efficiency.
- Dissolved salts may enter into the parts of other machinery, where steam is being used, thereby decreasing the life of the machinery.
- Actual height of the water column cannot be judged properly.

Foaming: It is the formation of continuous foams of bubbles due to presence of oil and alkalies in boiler feed water.

Causes of Foaming:

- It is caused by the presence of oil in boiler feed water.
- Presence of alkali in boiler feeds water.
- Oils and alkaline react to form soap which greatly lower the surface tension of water and thus increases the foaming tendency of the liquid.

Prevention of Priming

1. Controlling rapid change in steaming velocities.
2. The proper design of boilers.
3. Keeping the water level lower.
4. Filtration of the boiler water carried over to the boiler.
5. By blowing off sludge or scales from time to time.

Prevention of Foaming:

1. Adding antifoaming agents.
2. The removal of foaming agent (oil) from boiler water.
3. Adding castor oil which spread on the surface of water.
4. Blow down operation at appropriate time

References:

1. Engineering Chemistry,by S. Rattan

2. Engineering Chemistry, Wiley India

3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

Lime soda Process

Lime soda Process

In this process, soluble calcium magnesium and heavy metal salts in water are chemically converted in to insoluble precipitate of calcium carbonate and magnesium hydroxide by adding calculate amount of lime (calcium hydroxide) and soda (Na_2CO_3). The insoluble precipitate can be removed by filtration and sedimentation.

Lime removes temporary and permanent hardness, dissolved gases and free minerals acid from water.

Removal of temporary hardness:

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + H_2O$$

$$Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow Mg(OH)_2 \downarrow + CaCO_3 \downarrow + 2H_2O$$

Removal of permanent hardness:

$$\text{Permanent } MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2 \downarrow + CaCl_2$$

$$CaSO_4 + Ca(OH)_2 \longrightarrow \text{no reaction}$$

Removal of dissolved gases:

$$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 \downarrow + H_2O$$

$$H_2S + Ca(OH)_2 \rightarrow CaS \downarrow + 2\,H_2O$$

Removal of dissolved metals and free mineral acids:

$$FeSO_4 + Ca(OH)_2 \rightarrow Fe(OH)_2 \downarrow + CaSO_4$$

$$Al_2(SO_4)_3 + 3Ca(OH)_2 \rightarrow 2Al(OH)_3 \downarrow + 3CaSO_4$$

$$2HCl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O$$

$$H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$$

Removal of $NaHCO_3$

$$2NaHCO_3 + Ca(OH)_2 \longrightarrow Na_2CO_3 + CaCO_3 + 2H_2O$$

Coagulant is necessary because at room temperature precipitate is finally divided. But do not settle easily & cannot easily filtered.

$$NaAlO_2 + 2H_2O \longrightarrow NaOH + Al(OH)_3$$

On hydrolysis $NaAlO_2$ gives OH^- which is considered one equivalent of $Ca(OH)_2$.

Function of soda:

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + Na_2SO_4$$

$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + 2NaCl$$

Lime requirement for softening:

100 part of $CaCO_3$ equivalent hardness requires = 74 parts of $Ca(OH)_2$

Lime requirement = 74/100 [temporary Ca hardness + (2× temporary Mg hardness + permanent Mg hardness + (Salts of Al^{+3}+ Fe^{+2}) + CO_2 +H_2S +HCl +H_2SO_4 +HCO_3^- - $NaAlO_2$] all in terms of $CaCO_3$ equivalents.

Lime requirement = 74/100 [temporary Ca hardness + (2× temporary Mg hardness + permanent Mg hardness + (Salts of Al^{+3}+ Fe^{+2}) + CO_2 +H_2S +HCl +H_2SO_4 +HCO_3^- - $NaAlO_2$] X 100/ P.F. X Volume of water/10^6

Soda requirement for softening:

100 part of $CaCO_3$ equivalent hardness requires = 106 parts of **Na_2CO_3**

Soda required = 106/100 [permanent Ca + permanent Mg + salts of (Al^{+3}+ Fe^{+2}) + HCl + H_2SO_4 - HCO_3^-] all in terms of $CaCO_3$ equivalents.

Soda required = 106/100 [permanent Ca + permanent Mg + salts of (Al^{+3}+ Fe^{+2}) + HCl + H_2SO_4 - HCO_3^-] X 100/ P.F. X Volume of water/10^6

Lime soda process is two types.

Cold Lime –Soda (L-S) process: In this method, calculated quantities of lime and soda are mixed with water at room temperature. There are two kind of softener used for softening process-

1) **Intermittent type (Batch process)-** In this process hard water and calculated amount of lime and soda are mixed with the help of stirrers. Then soluble quantity of coagulants like sodium aluminate is added for setting the produced precipitate. The softened water is taken out through an exit pipe.

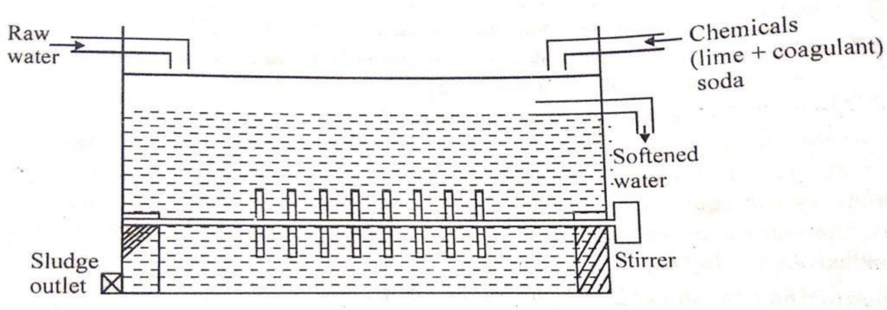
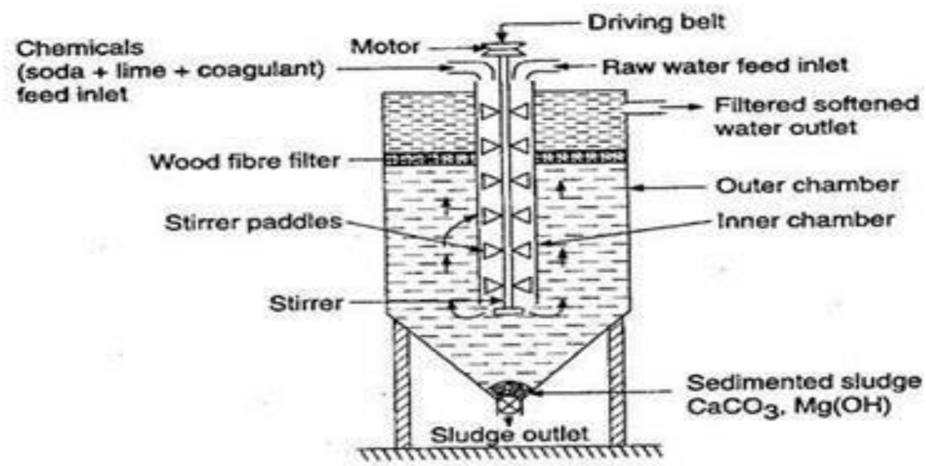


FIG.1 : BATCH PROCESS

2) **Continuous Softener**- This softener consists of a big steel tank with two chambers. The inner chamber is provided with a stirrer and outlet chamber has wood fiber filters to remove the fine particle of sludge. The raw water is passed into a inner tank with a static continuous flow, at the same time a calculated amount of chemical mixture of lime and soda are also added and then thoroughly mixed with a stirrer in this mixture tank after that water is allowed to settle. When the settlement becomes complete the water is passed by continuous up-flow process into another tank which is called filter bed. Then we get soften water.



Continuous Cold Lime –Soda Softener

The cold lime soda process is not suitable for all purposes. It is limited for those purposes where partially softened water needed because this process fails to precipitate completely all the hardness forming salts. The cold lime soda process is used for municipal water and cooling water softener systems.

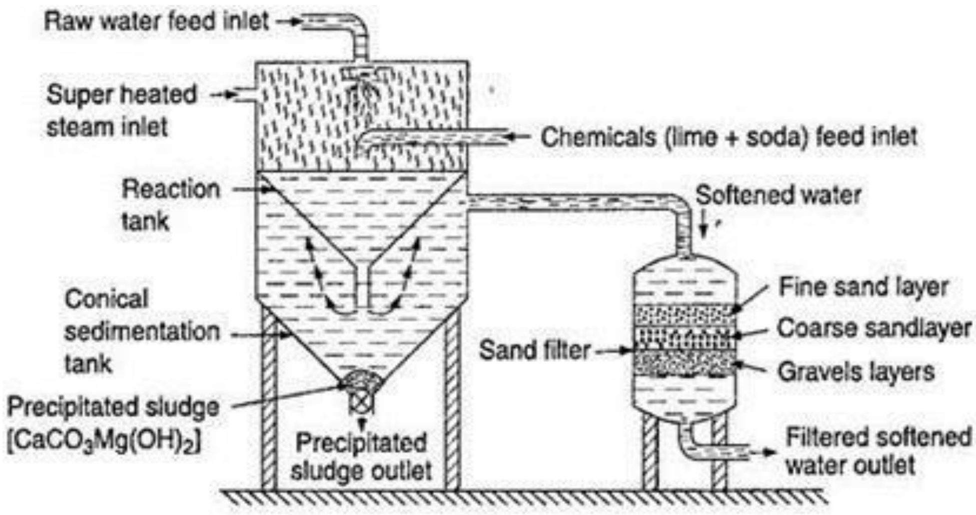
Hot lime soda process

In the hot lime soda process the reactions take place at higher temperature near about boiling point of water (80- 150ºC). The chemical mixing process is same as the cold lime soda process, but steam is applied in mixture tank. As a result precipitation becomes almost complete very quickly.

In this process continuous softener consist three parts-

a) **Reaction tank**- Raw water and chemicals are fed into this tank. Reactions starts and also complete in this tank
b) **Settling tank**- From the reaction tank, transfer into this tank where sludges settle down and removed by opening sludge valve.
c) **Sand filter-** Clear and soften water from settling tank passes to sand filter which remove

suspended particle and precipitate.



<u>Continuous Hot Lime –Soda Softener</u>

This process is more effective than cold lime soda process. The hot lime soda process water softener systems are exclusively used for boiler purpose.

Advantages:-

i) Lime –soda process is more economical.

ii) Less amount of coagulant is required.

iii) It increases pH value, there by corrosion is reduced.

vi) Pathogenic bacteria are reduced.

Disadvantages:-

i) The hardness is 50-60 ppm by cold process & about 15-30 ppm by hot process.

ii) Careful operation & skilled supervision are required.

iii) Disposal of large quantity of sludge formed during process is a problem.

Zeolite (Permutit) process

Zeolite (Permutit) process:

Zeolite : Zeolites are naturally occurring hydrated sodium alumina silicate minerals capable of exchanging its sodium ions with hardness producing ions in water.

These are represented as ***Na_2O. Al_2O_3. $xSiO_2$. yH_2O,***

where value of x = 2 to 10 and y = 2 to 6 .

Zeolites are also known as ***Permutits.***

Zeolites are two types.

1. **Natural Zeolites –** These are non-porous, amorphous and durable for example natrolite (**Na_2O. Al_2O_3. $4SiO_2$. $2H_2O$).
2. **Synthetic Zeolites-** These are porous and posses a crystalline structure. They are prepared by heating together sodium carbonates (Na_2CO_3), Alumina (Al_2O_3), and silica (SiO_2).

Process: Zeolites softener consists of a steel tank packed with a thick layer of permutit. When hard water is passed through a bed of active permutit then Ca and Mg salts present in it are removed in the form of insoluble zeolites and the resulting soft water is collected.

Zeolite simply represented as Na_2Z where Z is the insoluble radical frame work.

$$CaCl_2 + Na_2Z \rightarrow CaZ + 2NaCl$$

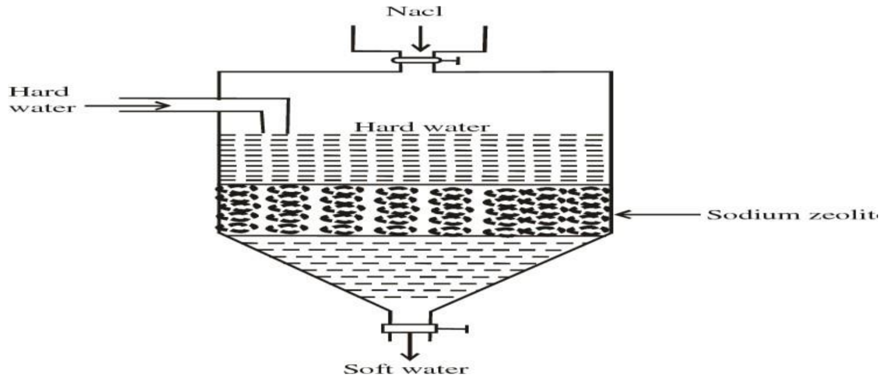
$$MgCl_2 + Na_2Z \rightarrow MgZ + 2NaCl$$

In this process water become free from Ca^{+2} and Mg^{+2} but get more concentrated with respect to sodium salt. Now, the bed loses its sodium exchange capacity and the Zeolite is said to be exhausted. The regeneration of zeolite is carried out by washing the bed with conc. solution of NaCl (brine solution).

$$CaZ + 2NaCl \rightarrow CaCl_2 + Na_2Z$$

$$MgZ + 2NaCl \rightarrow MgCl_2 + Na_2Z$$

(Exhausted zeolite) (*Regeneration zeolite)*



Zeolite Process for water Softening

Hardness of water can be calculated by the following formula-

Hardness = $50\times W\times V_2\times 10^3 / 58.5\times V_1$

Where V_1 = Total vol. of water softened

V_2 = Total vol. of NaCl used for regeneration

W = Amount of NaCl present in per litre water

This calculation used for the amount of NaCl consumed for regeneration of exhausted zeolite.

Advantage of Zeolite Process:

1) It is quite clean and rapid process which required less time for supporting**.**

2) The hardness completely removed and water of about 10ppm hardness is produced.

3) The equipment used is compact and occupies less space.

4) Impurities are not precipitate so there is no danger of sludge formation.

Disadvantage of Zeolite Process:

1) Softened water contains more sodium salts than lime soda process.

2) Highly turbid water cannot be treated efficiency by this method because fine impurities get deposited on the zeolite bed and create problems for its working.

3) The incoming water must be free from suspended matter otherwise zeolite gets clogged.

4) Lead cannot be removed.

L-9, Ion Exchange Process (Deionization or Demineralization)

Ion Exchange Process (Deionization or Demineralization): Ion exchange is a process of complete removal of all the ions present in water. Exchange of ions takes place between a solid (resins) and liquid (water) phase.

Ion exchange resins are organic, crosslinked, insoluble polymers having acidic or basic functional groups and are capable to exchange reversibly their H^+/OH^- ion with harness producing ions.

There are two general types of ion exchange resins:

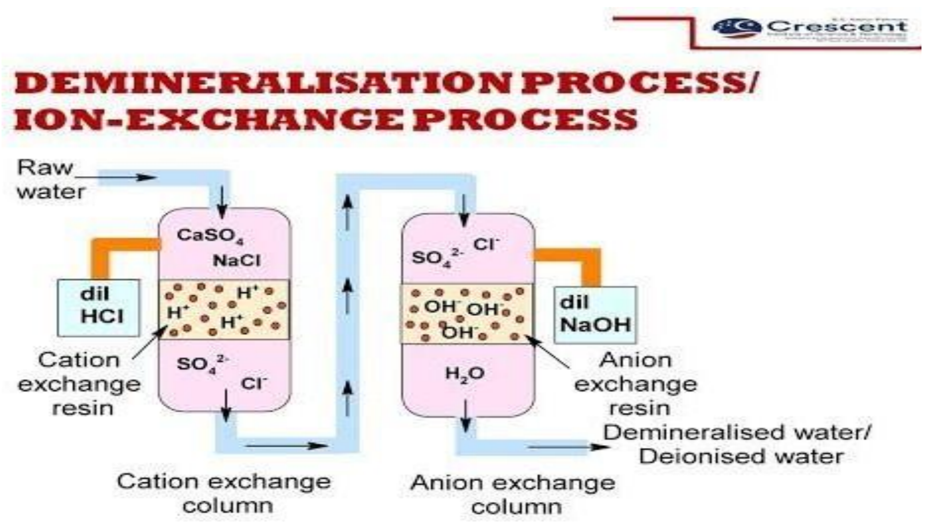
1) **Cation Exchange Resins**
2) **Anion Exchange Resins**

1) **Cations Exchange Resins (R^-H^+):** These resins exchange positive ions called **cation exchange resins**. A **cation** is an ion with a positive charge. These resins contain acidic groups like SO_3H, -COOH or –OH(phenolic) groups etc.

$$\mathbf{2R\text{-}COOH + Ca^{+2} \longrightarrow (RCOO)_2Ca + 2H^+}$$

2) **Anion Exchange Resins (R^+OH^-):** The resins that exchange negative ions called **anion exchange resins**. An anion is an ion with a negative charge. These resin containing functional groups like –NH_2, -$NH(CH_3)_2$ etc .

$$\mathbf{2R'(OH)_2 + SO_4^{-2} \longrightarrow R'SO_4 + 2OH^-}$$



Process-

The ion exchange softener consists of two tanks. The first tank contains cation exchanger and second contains anion exchanger.

The hard water first pass through cation exchange resin which exchanges its H^+ ions for Ca^{+2} and Mg^{+2} ions in water.

$$2R'H + CaCO_3 \longrightarrow R'_2Ca + H_2CO_3$$

$$2R'H + CaSO_4 \longrightarrow R'_2Ca + H_2SO_4$$

$$2R'H + MgCl_2 \longrightarrow R'_2Mg + 2HCl$$

($R' = RCOO^-, RSO_3^-, RO^-$)

Now water passing through anion exchanger tank. Hence the anion like SO_4^{-2}, Cl^- etc. present in water is exchanged by OH^- ions of the resin.

$$ROH + Cl^- \longrightarrow RCl + OH^-$$

$$2ROH + SO_4^{-2} \longrightarrow R_2SO_4 + 2OH-$$

Thus water coming from after anion exchanger free from cations and anions which are responsible for hardness.

H^+ ions released from cation exchanger and OH^- ions released from anion exchanger combine to form unionized water.

$$H^+ + OH^- \longrightarrow H_2O$$

Regeneration

The "reactivation" process is called regeneration and is carried out using a strong acid for the cation (as a source of hydronium ions) and liquid caustic (sodium hydroxide) as a source of hydroxyl ions for the anion.

$$R'_2Ca + H_2SO_4 \longrightarrow 2R'H + CaSO_4$$

$$R'SO_4 + NaOH \longrightarrow R'(OH)_2 + Na_2SO_4$$

Advantage of Ion Exchange Process:

1. Highly acidic or alkaline water can be treated by this process.
2. This produces water of very low hardness nearly 2 ppm.

Disadvantage of Ion Exchange Process:

1. The equipment is costly and more expensive chemical are required.
2. If water contains turbidity then the output of the process is reduced.

References:

1. Engineering Chemistry,by S. Rattan

2. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

L-10 Numerical problems based on Hardness, EDTA, Lime- Soda and Zeolite

Calculate the temporary and total hardness of a water sample containing $Mg(HCO_3)_2$= 74mg/L, $Ca(HCO_3)_2$= 160mg/L, $MgCl_2$= 95mg/L, $CaSO_4$=132mg/L

Solution: calculation of $CaCO_3$ equivalents:

Constituent	Multiplication factor	$CaCO_3$ equivalent
$Mg(HCO_3)_2$= 74mg/L	100/146	74X100/146= 50.68mg/L
$Ca(HCO_3)_2$= 160mg/L	100/162	160X100/162=98.76mg/L
$MgCl_2$= 95mg/L	100/95	95X100/95= 100mg/L
$CaSO_4$=132mg/L	100/136	132X100/136= 97.05mg/L

∴ Temporary hardness of water due to $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$ =

=149.44 ppm

Total hardness of water = 50.68+98.76+100+97.05 =346.49 mg/L or ppm

A 500 ml SHW containing 0.5 g $CaCO_3$. 50ml of the Standard water solution required 48ml of EDTA solution for titration. 50ml of hard water sample required 15ml of EDTA and after boiling and filtering required 10ml of EDTA solution. Calculate the hardness

Solution:

500ml of SHW = 0.5g or 500mg $CaCO_3$ eq

∴1ml SHW= 1mg $CaCO_3$ eq

Now 48ml of EDTA solution = 50/48 mg $CaCO_3$ eq

∴1ml of EDTA solution= 50/48 mg $CaCO_3$ eq

Calculation of the total hardness of water:

50ml hard water = 15ml EDTA = 15x50/48 mg of $CaCO_3$ eq

∴1000ml of hard water= 15.625x1000/50 = 312.5 mg $CaCO_3$ eq

Hence total hardness= 312.5mg/L or 312.5 ppm.

A 10,000 litre of water containing the following salts dissolved per litre. $MgSO_4$=12 mg, $CaCl_2$=11.1 mg, $NaHCO_3$=7.25 mg, $CaCO_3$=50 mg silica=10 mg, NaCl= 29mg Calculate the amount of lime and soda required.

Solution: conversion of the hardness in terms of $CaCO_3$ equivalents

Constituent	**Multiplication factor**	**$CaCO_3$ equivalent**
$CaCO_3$=50 mg	100/100	50x100/100=50 mg/L
$CaCl_2$=11.1 mg	100/111	11.1x100/111=10 mg/L
$MgSO_4$=12 mg	100/120	12x100/120=10 mg/L
$NaHCO_3$=7.25 mg	100/84	7.25x100/84=8.63 mg/L

Lime required= 74 [$CaCO_3$+1/2 $NaHCO_3$+$MgSO_4$]/100

= 74/100 [50+4.3+10]

= 47.58 mg/L

Amount of lime required for 10,000L of water

= 47.58x10,000 = 4,75,800 mg/L

= 0.4758 Kg/L

Amount of Soda required=106 [$CaCl_2$+$MgSO_4$-1/2 $NaHCO_3$]/100

= 106 [10+10-4.3] /100

= 16.64mg/L

Soda required for 10,000L of water= 16.64x10,000 = 166400 mg/L

= 0.1664 Kg/L

References:

1. Engineering Chemistry,by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram